SYNOPSIS Date: 02-09-2010

FAMILY OF LINES THROUGH THE INTERSECTION OF TWO GIVEN LINES

THEOREM: Prove that the equation of the family of lines passing through the intersection of the lines $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$ is

$(a_1x+b_1y+c_1)+\lambda(a_2x+b_2y+c_2)=0$, where λ is a parameter.

PROOF : The equations of the lines are

$a_1x+b_1y+c_1=0$ ....(i)

and $a_2x+b_2y+c_2=0$ .....(ii)

Let (α,β) be the point of intersection of the lines (i) and (ii). then,

$a_1\alpha+b_1\beta+c_1=0$ .....(iii)

and , $a_2\alpha+b_2\beta+c_2=0$ ......(iv)

Now, consider the equation

$(a_1x+b_1y+c_1)+\lambda(a_2x+b_2y+c_2)=0$ ......(v)

Clearly, this is a first degree equation in x and y. So it represents a straight line.

We have, $(a_1\alpha+b_1\beta+c_1)+\lambda(a_2\alpha+b_2\beta+c_2)$ = $0+\lambda 0=0$ [Using (iii) and (iv)]

So, (α,β) lies on the line given in (v).

Hence, (v) represents family lines through the point of intersection of (i) and (ii). Thus, the family of straight lines through the intersection of lines

$L_1=a_1x+b_1y+c_1=0$ and $L_2=a_2x+b_2y+c_2=0$ is

$(a_1x+b_1y+c_1)+\lambda(a_2x+b_2y+c_2)=0$

i.e., $L_1+\lambda L_2=0$.

REMARK: The equation $L_1+\lambda L_2=0$ represents a line passing through the intersection of the lines L_1=0 and L_2=0 which is a fixed point. Hence, $L_1+\lambda L_2=0$ represents a family of straight lines, for different values of λ, which pass through a fixed - point.

Example: Find the equation of the straight line which passes through the point (2,-3) and the point of intersection of the lines x + y + 4 = 0 and x - y - 8 = 0.

Solution Any line through the intersection of the lines x + y + 4 = 0 and 3 x - y - 8 = 0. has the equation

$(x+y+4)+\lambda(3x-y-8)=0$ ....(i)

this will pass through (2,-3) if (2 - 3 + 4) + λ(6 + 3 - 8) =0

$\Rightarrow 3+\lambda=0 \Rightarrow \lambda=-3$.

Putting the values of λ in 9i), the required line is 2x – y – 7 = 0

ALTER : Solving the equations x + y + 4 = 0 and 3x- y - 8 =0 by cross - multiplication, we get x = 1, y = - 5.

So the two lines intersect at the point (1,-5). Hence, the required line passes through (2,-3) and (1,-5) and so its equation is .

$$y+3=-\frac{5+3}{1-2}(x-2) \Rightarrow 2x-y-7=0.$$

EQUATION OF THE BISECTORS.

THEOREM. Prove that the equation of the bisectors of the angles between the lines.

$$a_1x+b_1y+c_1=0 \qquad \ldots.(i)$$

and, $a_2x+b_2y+c_2=0 \qquad \ldots.(ii)$

are given by $\dfrac{a_1x+by+c_1}{\sqrt{a_1^2+b_1^2}}=\pm\dfrac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$

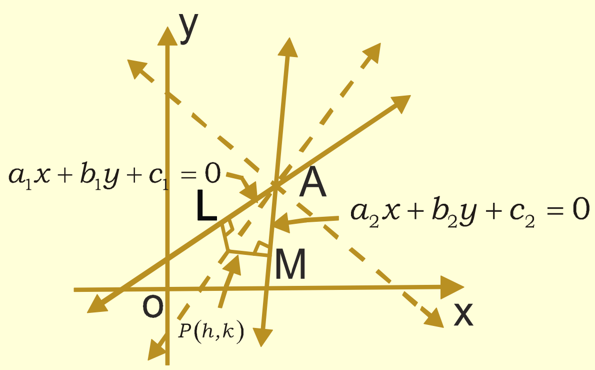
PROOF: The bisectors of the angles between two straight lines are the locus of a point which is equidistant from the two lines. So let P(h,k) be a point equidistant from lines (i) and (ii) . Then

$$PL = PM \Rightarrow \left|\frac{a_1h+b_1k+c_1}{\sqrt{a_1^2+b_1^2}}\right|=\left|\frac{a_2h+b_2k+c_2}{\sqrt{a_2^2+b_2^2}}\right|$$

$\therefore$ Locus of (h,k) i.e., the equations of the bisectors are

$$\frac{|a_1x+b_1y+c_1|}{\sqrt{a_1^2+b_1^2}}=\frac{|a_2x+b_2y+c_2|}{\sqrt{a_2^2+b_2^2}}$$

$$\Rightarrow \frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}=\pm\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$$



$[\because |a|=|b| \Leftrightarrow a=\pm b]$

These are the required equations of the bisectors. Q.E.D.

ILLUSTRATION 1. Find the equations of the bisectors of the angles between the straight lines 3x - 4y + 7 =0 and 12x - 5y -8 =0.

SOLUTION The equations of the bisectors of the angles between 3x - 4y + 7 =0 and 12x - 5y -8 =0 are

$$\frac{3x-4y+7}{\sqrt{3^2+(-4)^2}}=\pm\frac{12x-5y-8}{\sqrt{12^2+(-5)^2}} \quad \text{or,} \quad \frac{3x-4y+7}{5}=\pm\frac{12x-5y-8}{13}$$

or , 39x - 52y + 91 = $\pm$ (60x - 25y -8)

Taking the positive sign, we get 21x + 27y - 131 = 0 as one bisector.

Taking the negative sign, we get 99x - 77y + 51 = 0 as the other bisector.

ACUTE AND OBTUSE ANGLE BISECTORS

Let the equations of the two line be

$$a_1x+b_1y+c_1=0 \qquad \text{....(i)}$$

and, $a_2x+b_2y+c_2=0$ ....(ii)

where $c_1>0, c_2>0$ Consider the bisector

$$\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}=\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}} \qquad \text{....(iii)}$$

Let, $m_1, m_2, and\, m_3$ be the slopes of (i), (ii), (iii) respectively.

Then, $m_1=\dfrac{-a_1\sqrt{a_2^2+b_2^2}-a_2\sqrt{a_1^2+b_1^2}}{b_1\sqrt{a_2^2+b_2^2}-b_2\sqrt{a_1^2+b_1^2}}$

Let θ be the angle between (i) and (ii). Then,

$$tan\,\theta=\frac{\dfrac{-a_1}{b_1}+\dfrac{a_1\sqrt{a_2^2+b_2^2}-a_2\sqrt{a_1^2+b_1^2}}{b_1\sqrt{a_2^2+b_2^2}-b_2\sqrt{a_1^2+b_1^2}}}{1+\dfrac{a_1}{b_1}\cdot\dfrac{a_1\sqrt{a_2^2+b_2^2}-b_2\sqrt{a_1^2+b_1^2}}{b_1\sqrt{a_2^2+b_2^2}-b_2\sqrt{a_1^2+b_1^2}}}$$

$$\Rightarrow tan\,\theta=\frac{(a_1b_2-a_2b_1)\sqrt{a_1^2+b_1^2}}{(a_1^2+b_1^2)\sqrt{a_2^2+b_2^2}-(a_1a_2+b_1b_2)\sqrt{a_1^2+b_1^2}}$$

$$\Rightarrow tan\,\theta=\frac{(a_1b_2-a_2b_1)}{(a_1^2+b_1^2)\sqrt{a_2^2+b_2^2}-(a_1a_2+b_1b_2)}$$

$$\Rightarrow tan\,\theta = \frac{a_1b_2 - a_2b_1}{\sqrt{(a_1b_2 - a_2b_1)^2 + (a_1a_2 + b_1b_2)^2} - (a_1a_2 + b_1b_2)}$$

$$\Rightarrow tan\,\theta = \frac{u}{\sqrt{u^2 + v^2} - v'}$$ where $u = a_1b_2 - a_2b_1$ and $v = a_1a_2 + b_1b_2$

$$\Rightarrow |tan\,\theta| = \frac{|u|}{\sqrt{u^2 + v^2} - v}$$

Now, two case arise :

CASEI : When $v = a_1a_2 + b_1b_2 < 0$:

in this case, we have $v < 0 \Rightarrow -v < 0$ Also, $\sqrt{u^2 + v^2} > \sqrt{u^2} = |u|$

$\therefore \sqrt{u^2 + v^2} - v > |u| \Rightarrow |tan\,\theta| < 1$

$\Rightarrow$ Equation (iii) gives the bisector of the acute angle between the given lines.

CASEII: When $v = a_1a_2 + b_1b_2 > 0$

We know that $\sqrt{u^2 + v^2} \le |u| + |v| \Rightarrow \sqrt{u^2 + v^2} - v|u|$ $[\because |v| - v]$

$$\Rightarrow \frac{|u|}{\sqrt{u^2 + v^2} - v} > 1 \Rightarrow |tan\,\theta| > 1$$

$\Rightarrow$ Equation (iii) is the bisector of the obtuse angle between the lines (i) and (ii).

Thus, if c_1, c_2 are positive and $a_1a_2 + b_1b_2 < 0$, then $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$

is the bisector of acute angle between the lines and consequently.

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = -\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

is the bisector of the obtuse angle between the lines. Also, if c_1, c_2 are positive

and $a_1a_2 + b_1b_2 > 0$, then $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$

is the bisector of the obtuse angle between the lines and consequently

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = -\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

is the bisector of acute between them. From the above discussion, we obtain the following algorithm to find the bisectors of acute and obtuse angles between the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$.

ALGORITHM

Let the equations of the two lines be

$a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$

STEP I : See whether the constant terms c_1 and c_2 in the two equations are positive or not. If not, then multiply both sides of the given equations by - 1 to make the constnat terms positive.

STEP II. Determine the sign of the expression $a_1a_2 + b_1b_2$.

STEP III. If $a_1a_2 + b_1b_2 > 0$, then the bisector corresponding to " + " sign gives the obtuse angle bisector and the bisector corresponding to "—" sign is the bisector of acute between the line i.e.,

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = +\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}} \quad \text{and} \quad \frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = -\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$ are the

bisectors of obtuse and acute angles respectively.

If $a_1a_2 + b_1b_2 < 0$, then the bisector corresponding to "+" and "—" sign give the acute and obtuse angle bisectors respectively i.e.,

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = +\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}} \quad \text{and} \quad \frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = -\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

are the bisectors of acute and obtuse angles respectively.

EXAMPLE : Find the coordinates of those points on the line 3x + 2y = 5 which are equidistant from the lines 4x + 3y -7 =0 and 2y - 5 = 0.

Solution: We know that the bisectors of the angles between two lines contain all those points which are equidistant from the lines. So, the required point lies on the bisectors of the angles between the given lines.

The equations of the bisectors are given by

$$\frac{4x + 3y - 7}{\sqrt{4^2 + 3^2}} = \pm\frac{2y - 5}{\sqrt{6^2 + 2^2}} \Rightarrow \frac{4x + 3y - 7}{5} = \pm\frac{2y - 5}{2}$$

$\Rightarrow 8x - 4y + 11 = 0$ and 8x +16y -39 = 0

Let (h,k) be the coordinates of the required point. Then ,

8h + 16h + 11=0 .....(i)

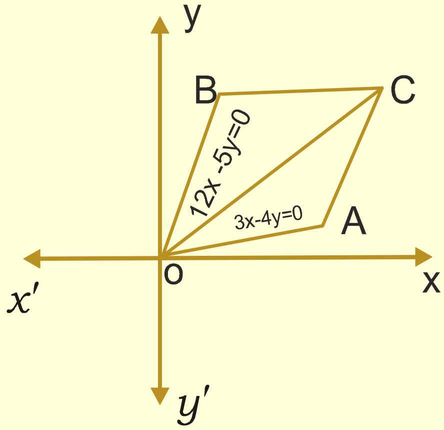
and 8h + 16k -39 =0 ....(ii)

It is given that (h,k) lies on 3x + 2y = 5. Therefore, 3h+ 2 k = 5 ....(iii)

Solving (iii) with (i) and (ii) respectively, we obtain the coordinates of the required points as (-1/14, 73 /28) and (1/16,77/32) .

EXAMPLE : Two sides of a rhombus, lying in the first - quadrent, are given by 3x - 4y = 0 and 12x -5y =0. If the length of the longer diagonal is 12, find the equations of the other two sides of the rhombus.

Solution : The equations of the sides OA and OB are 3x - 4y = 0 and 12x - 5y = 0 respectively. Clearly, OC is the longer diagonal of the rhombus OACB. It is one of the bisectors of the angles between OA and OB. The equations of the bisectors are given by



$$\frac{3x-4y}{5}=\pm\frac{12x-5y}{13}\Rightarrow 9x-7y=0 \text{ and } 7x+9y=0$$

So, the equation of OC is 9x - 7y = 0. Note that 7x + 9y = 0 cannot be the equation of OC as it makes an obtuse angle with OX.

The equation of AC can also be written in the distance form as $\frac{x}{\frac{7}{\sqrt{130}}}=\frac{y}{\frac{9}{\sqrt{130}}}$

therefore, the coordinates of C are given by

$$\frac{x}{\frac{7}{\sqrt{130}}}=\frac{y}{\frac{9}{\sqrt{130}}}=12 \qquad [\because \text{ OC} = 12 \text{ (given)}]$$

$$\Rightarrow x=\frac{84}{\sqrt{130}},y=\frac{108}{\sqrt{130}}$$

thus, the coordinates of C are $\left(\frac{84}{\sqrt{130}},\frac{108}{\sqrt{130}}\right)$.

Clearly AC and BC are lines passing through C and parallel to the sides OB and OA respectively. So, their equations are

$$y-\frac{108}{\sqrt{130}}=\frac{12}{5}\left(x-\frac{84}{\sqrt{130}}\right) and\ y-\frac{108}{\sqrt{130}}=\frac{3}{4}\left(x-\frac{84}{\sqrt{130}}\right)$$ respectively.

WORKSHEET - 13 Date: 02-09-2010

LEVEL - 1

SINGLE CORRECT CHOICE TYPE:

1. The straight line (2 + 3k)x + (3 – 5k)y + (4 – 13k) = 0 for any value of k passes through a point

   1) (1, –2) 2) (0, 0) 3) (2, 1) 4) (1, 2)

2. The point of concurrecy of the family of the straight lines

   (k + 1)x +(k + 2)y + 5 = 0 is

   1) (5, 4) 2) (5, –5) 3) (0, 1) 4) (1, 0)

3. The equation of one of the bisectors bisecting the angles between the lines 3x – 4y + 7 = 0 and 12x + 5y – 2 = 0 is

   1) 11x + 3y + 9 = 0 2) 11x – 3y + 9 = 0 3) 11x – 3y – 9 = 0 4) None

MULTI CORRECT CHOICE TYPE:

4. Two sides of an isosceles triangle are given by the equation 7x – y + 3= 0 and x + y – 3 = 0 the slope of third side is

   1) – 3 2) $1/3$ 3) $-1/3$ 4) 3

REASONING TYPE:

5. *Statement I :* One of the bisectoes of the angle between the lines $a(x-1)^2+2h(x-1)(y-2)+b(y-2)^2=0$ is x + 2y – 5 = 0, then other bisector is 2x – y = 0.

   Statement I: The angular bisectors of lines will always be mutually perpendicular.

   1. Both Statements are true, Statement II is the correct explanation of Statement I.
   2. Both Statements are true, Statement II is not correct explanation of Statement I.
   3. Statement I is true, Statement II is false.
   4. Statement I is false, Statement II is true.

6. *Statement I:* If a,b,c are in A.P ,then $ax+by+c=0$ passes through the point (1,-2)

   Statement II : The equation of the family of lines passing through the intersection of lines $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$ is $(a_1x+b_1y+c_1)+\lambda(a_2x+b_2y+c_2)=0$,where λ is parameter.

   1. Both Statements are true, Statement II is the correct explanation of Statement I.
   2. Both Statements are true, Statement II is not correct explanation of Statement I.
   3. Statement I is true, Statement II is false.
   4. Statement I is false, Statement II is true.

COMPREHENSION TYPE:

Consider the lines $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$. If $a_1a_2+b_1b_2>0$ and $c_1c_2>0$ the equation of the bisector of the acute angle between the lines is

$$\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}=-\frac{(a_2x+b_2y+c_2)}{\sqrt{a_1^2+b^2}}$$

If $a_1a_2+b_1b_2<0$ and $c_1c_2>0$, the eqn of the acute angular bisector between the given lines is $\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}=\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$

If c_1,c_2 have the same signs, eqn. of the bisector containing origin is

$$\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}=\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$$

Based on the above information, answer the following questions.

7. The acute angular bisector between the lines $3x+4y+3=0$ and $4x+3y+8=0$ is

   1) $7x+7y-11=0$ 2) $7x-7y+11=0$ 3) $7x+7y+11=0$ 4) $7x+7y+4=0$

8. The equation of obtuse angular bisector between the lines x - y + 2 = 0 and 7x + y + 1 =0 is

   1) $6x+2y+1=0$ 2) 6x + 2y - 1=0 3) 2x +6y - 1=0 4) 2x + 6y +1 = 0

9. The equation of bisector containing origin between the lines 3x - 4y +2 = 0 and 12x - 5y +2=0 is

   1) 21x +27y-16=0 2) 21x-27y +6=0 3) 21x+27y+16=0 4) 21x-27y-16=0

MATRIX MATCH TYPE:

10.

Column-I	**Column-II**
System of lines	***Point of concurrence***
a. $(3k+1)x-(2k+3)y+(9-k)=0$	p) (-2,1)
b. $(a+2b)x+(a-b)y+(a+5b)=0$	q) (3,4)
c. $(2x+3y+1)+k(3x-2y-5)=0$	r) (2,2)
d. $a(x+y-4)+b(2x-y-2)=0$	s) (1,-1)

LEVEL - 2

SINGLE CORRECT CHOICE TYPE:

11. The equation of the straight line passing through the point of entersection of the lines x + y + 1 = 0 and 2x – y + 5 = 0 and containing the point (5, – 2) is

    1) 9x + 21y + 3 = 0 2) 3x – 7y – 120 3) 9x – 21y – 3 = 0 4) 3x + 7y – 1 = 0

MULTI CORRECT CHOICE TYPE:

12. Given three striaght lines $2x + 11y - 5 = 0$, $24x + 7y - 20 = 0$ and $4x - 3y - 2 = 0$. Then,

    1) they form a triangle
    2) they are concurrent
    3) one line bisects the angle between the other two
    4) two of them are parallel.

LEVEL - 3

13. The point of concurrecy of the family of the straight lines $(2 + 5k) x - 3(1 + 2k)y + (2 - k) = 0$ is

    1) 5, 4 2) (0, 2) 3) (3, 2) 4) (5, 1)

MULTI CORRECT CHOICE TYPE:

14. Sides of a rhombus are parallel to the lines $x + y - 1 = 0$ and $7x - y - 5 = 0$. It is given that diagonals of the rhombus intersect at (1, 3) and one vertex, 'A' of the rhombus lies on the line $y = 2x$. Then the coordinates of the vertex A are

    1) (8/5, 16/5) 2) (7/15, 14/15) 3) (6/5, 12/5) 4) (4/15, 8/15)

LEVEL - 4

SINGLE CORRECT CHOICE TYPE:

15. The lines $x + y - 1$, $x - y - 1 = 0$ and $x - 3y + 3 = 0$

    1) form an equilateral triangle
    2) form isoscelels triangle
    3) are concurrect
    4) none

16. Find the equation of the bisector the obtuse angle between the line $3x - 4y + 7 = 0$ and $12x + 5y - 2 = 0$.

LEVEL - 5

COMPREHENSION TYPE:

For the straight lines $4x + 3y - 6 = 0$ and $5x + 12y + 9 = 0$

17. The equation of the bisector of the obtuse angle between given equations is

    1) $9x - 7y - 41 = 0$ 2) $7x - 9y - 3 = 0$ 3) $7x + 9y - 3 = 0$ 4) $9x + 7y + 41 = 0$

18. The equation of the acute angle between given equations is

    1) $9x - 7y - 41 = 0$ 2) $7x - 9y - 3 = 0$ 3) $7x + 9y - 3 = 0$ 4) None of these

19. The equation of the bisector of the angle which contains origin is

    1) $9x - 7y - 41 = 0$ 2) $7x - 9y - 3 = 0$ 3) $7x + 9y - 3 = 0$ 4) $9x + 7y + 41 = 0$

20. The equation of the bisector of the angle which contains (1, 2) is

    1) $9x - 7y - 41 = 0$ 2) $7x - 9y - 3 = 0$ 3) $7x + 9y - 3 = 0$ 4) $9x + 7y + 41 = 0$

SYNOPSIS Date: 03-09-2010

DEFINITION OF LOCUS

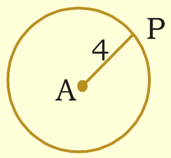
Let us call a set of geometric conditions 'consistent' if there is atleast one point satisfying that set of conditions. For example when A = (1,0) and B= (3,0) the condition 'the sum of distances of a point P from A and B is equal to 2' is consistent, whereas the condition 'the sum of distances of a point Q from A and B is equal to 1' is not consistent, becuase there is no point Q such that QA +QB = 1 (since AB = 2).

By the locus of a point we mean the set of all positions that it can take when it is subjected to certain consistent geometric conditions.

Example:

Locus of a point whose distance from a given point A is 4 units.

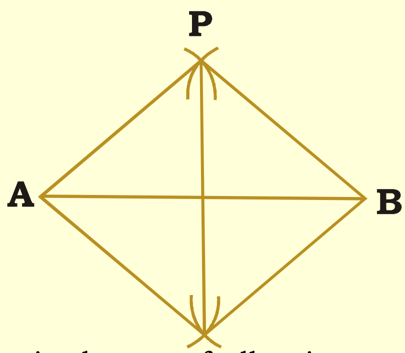
Any point which is at a distance of 4 units from A lies on the circle of radius 4 units with centre A. Obviously, every point on the circle is at a distance of 4 units from A. Hence, the set of all points on the circle is the locus in this example



Example

The locus of a point equidistant from two given points A,B. The mid - point of AB is a point on the locus. All the points equidistant from A and B are obtained by drawing arcs using a compass. The set of all points, thus obtained is the locus.

i.e., locus - { P | AP = BP } .



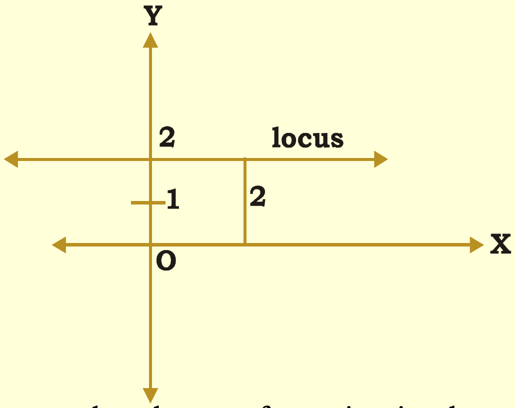
In this example, the locus is the set of all points on the perpendicualr bisector of $\overline{AB}$ (since, AP = BP $\Leftrightarrow$ P lies on the perpendicular bisector).

Example:

Locus of a point whocs distance above X - axis is 2 units.

It can be easily seen that any point satisfying the given conditions lies on the straight line above the X -axis, parallel to it at distance of 2 units and any point on this line satisfies the given conditions.

So, locus = $\{ (x,2) \mid x \in R\}$



In view of the above examples, locus of a point in the plance is generally a curve on the plane. for simplicity, we call that curve itself as locus. The locus in example 1 is a circle, whereas it is a straight line in examples 2 and 3.

Equation of Locus

It is clear that, every point on the locus satisfies the given conditions and every point which satisfies the given conditions lies on the locus .

By the equation of a locus we mean an algebraic description of the locus. It is obtained by translating the geometric conditions satisfied by the points on the locus, into equivalent algebraic conditions.

Algebraic descriptions give rise to algebraic equations which some times contair more than what is required by the geometric conditions. Thus locus may be a part of the curve represented by the algebraic equation. Usually we call this algebraic equation as the equation of locus. However, to get the full description of the locus, the exact part of the eurve, points of which satisfy the given geometric description, must be specified.

Solved problems:

Problem.

Find the equation of the locus of a point which is at a distance of 5 units from A(4,-3).

Solution.

A(4,-3) is a fixed point . Let P(x,y) be a point on the locus.

The geometric condition to be satisfied by P is that

AP = 5 units. ....(1)

Expressing this condition algebraically , we get $\sqrt{(x-4)^2+(y+3)^2}=5$

i.e., $x^2-8x+16+y^2+6y+9=25$

i.e., $x^2+y^2-8x+6y=0$ ....(2)

Let $Q(x_1,y_2)$ satisfy (2). Then , $x_1^2+y_1^2-8x_1+6y_1=0$

Now the distance between A and Q is AQ = $\sqrt{(x_1-4)^2+(y_1+3)^2}$

$\therefore \quad AQ^2 = x_1^2 - 8x_1 + 16 + y_1^2 + 6y_1 + 9$

$= \left(x_1^2 + y_1^2 - 8x_1 + 6y_1\right) + 25 = 25$ (by using (3))

Hence AQ = 5 units. This means that $Q\left(x_1, y_1\right)$ satisfies the geometric condition(1). Therefore, equation of the locus is $x^2 + y^2 - 8x + 6y = 0$

Problem.

Find the locus of the third vertex of a right angled triangle, the ends of whose hypotenuse are 94,0) and (0,4) .

Solution.

Let A= (4,0) and B = (0,4). Let P (x,y) be a point such that PA and PB are perpendicular. Then $PA^2 + PB^2 = AB^2$ and P,A,B are non collinear ....(1).

i.e., $\left(x-4\right)^2 + y^2 + x^2 + \left(y-4\right)^2 = 16 + 16, P \neq A \; and \; p \neq B$

i.e., $2x^2 + 2y^2 - 8x - 8y = 0,\; P \neq A$ and $P \neq B$

or $x^2 + y^2 - 4x - 4y = 0, \left(x, y\right) \neq \left(4, 0\right)$ and (x, y) $\neq$ (0,4) ......(2)

Let $Q\left(x_1, y_1\right)$ satisfy (2) and Q is different from A and B.

Then $x_1^2 + y_1^2 - 4x_1 - 4y_1 = 0, \left(x_1, y_1\right) \neq \left(4, 0\right)$ and $\left(x_1, y_1\right) \neq \left(0, 4\right)$ .....(3)

Now $AQ^2 + QB^2 = \left(x_1 - 4\right)^2 + y_1^2 + x_1^2 + \left(y_1 - 4\right)^2$

$= x_1^2 + 16 - 8x_1 + y_1^2 + x_1^2 + y_1^2 + 16 - 8y_1 = 2\left(x_1^2 + y_1^2 - 4x_1 - 4y_1\right) + 32$

= 32 (by using (3)) = AB^2.

Hence, $QA^2 + QB^2 = AB^2, Q \neq A$ and $Q \neq B$

This means that $Q\left(x_1, y_1\right)$ satisfies (1). Therefore, equation of the locus is (2) , which is the circle except A and B.

WORKSHEET - 14 Date: 03-09-2010

LEVEL-1

SINGLE CORRECT CHOICE TYPE:

1. The locus of the point which is at a distance 5 unit from (-2,3) is

1) $x^2 - y^2 + 4x - 6y + 12 = 0$ 2) $x^2 + y^2 + 4x - 6y - 12 = 0$

3) $x^2 - y^2 + 4x - 6y - 12 = 0$ 4) $x^2 + y^2 + 4x - 6y + 12 = 0$

2. The locus of the point which is at a distance 5 units from the origin is

1) $x^2 + y^2 = 25$ 2) $x^2 + y^2 = 19$ 3) $x^2 + y^2 = 32$ 4) $x^2 + y^2 = 29$

3. The locus of the point, for which the sum of the squares of distances from the coordinate axes is 25 is

1) $x^2+y^2=25$ 2) $x^2+y^2=19$ 3) $x^2+y^2=32$ 4) $x^2+y^2=29$

MULTI CORRECT CHOICE TYPE:

4. The locus of the point which is at a distance 2 unit from y - axis is

1) $x^2+4=0$ 2) $x^2-4=0$ 3) $(x-2)(x+2)=0$ 4) $x^2+2=0$

REASONING TYPE:

5. *Statement I :* If A(a,0), B(-a,0), then the locus of the point P such that $PA^2+PB^2=2c^2$ is $x^2+y^2+a^2-c^2=0$

Statement II : The distance between the points $A(x_1,y_1)$ and $B(x_2,y_2)$ is

$$AB=\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$$

1) Both Statements are true, Statement II is the correct explanation of Statement I.
2) Both Statements are true, Statement II is not correct explanation of Statement I.
3) Statement I is true, Statement II is false.
4) Statement I is false, Statement II is true.

COMPREHENSION TYPE:

The area of the triangle formed by the points $(x_1,y_1),(x_2,y_2),(x_3,y_3)$ is

$$\frac{1}{2}\begin{vmatrix} x_1-x_2 & x_1-x_3 \\ y_1-y_2 & y_1-y_3 \end{vmatrix}$$

6. A(2,3), B(-3,4) are two points. If a point P moves such that the area of Δ PAB is 8.5 sq. unit, then the locus of P is .

1) $x^2+10xy+25y^2-34x-170y=0$ 2) $x^2+10xy-25y^2-34x-170y=0$

3) $x^2-10xy+25y^2-34x+170y=0$ 4) $x^2-10xy-25y^2+34x-170y=0$

7. The locus of p such that area of Δ PAB is 12 square units where A=(2,3) and B=(-4,5) is

1) $x^2+6xy+9y^2+22x+66y+23=0$ 2) $x^2-6xy+9y^2+22x+66y+23=0$

3) $x^2+6xy+9y^2-22x-66y-23=0$ 4) $x^2-6xy+9y^2-22x-66y-23=0$

8. $O(0,0),A(4,0),B(0,6)$ are three points. If P is a point such that area of Δ POB is twice the area of Δ POA, then the locus of P is .

1) $4x^2-6y^2=0$ 2) $3x^2-4y^2=0$ 3) $9x^2-16y^2=0$ 4) $4x^2-9y^2=0$

MATRIX MATCH TYPE:

9. **Column - I** **Column - II**

Column - I	Column - II
a) The locus of the point which is at a distance 5 units from x - axis is	p) x = 0
b) The locus of the point which is equidistance to the coordinate axes is	q) 2x - y + 4 =0
c) The equation to the locus of points equidistance from the points (2,3), (-2,5) is	r) $x^2 - y^2 = 0$
d) The equation to the locus of points equidistance from the points (-3,4), (3,4) is	s) $y^2 - 25$

LEVEL-2

SINGLE CORRECT CHOICE TYPE:

10. The locus of the point whose distances to the coordinate axes are in the ratio 2 : 3 is

1) $3x^2 - 4y^2 = 0$ 2) $4x^2 - 3y^2 = 0$ 3) $4x^2 - 16y^2 = 0$ 4) $4x^2 - 9y^2 = 0$

11. The equation to the locus of a point P for which the distance from P to (4,0) is double the distance from P to x - axis is

1) $x^2 + 3y^2 + 8x + 16 = 0$ 2) $x^2 + 3y^2 - 8x - 16 = 0$

3) $x^2 - 3y^2 + 8x - 16 = 0$ 4) $x^2 - 3y^2 - 8x + 16 = 0$

REASONING TYPE:

12. *Statement I :* The equation to the locus of points which are equidistant from the points (-3,2), (0,4) is 6x + 4y -3=0.

Statement II : The locus of points which are equidistant to A,B is perpendicular bisector of AB

1) Both Statements are true, Statement II is the correct explanation of Statement I.
2) Both Statements are true, Statement II is not correct explanation of Statement I.
3) Statement I is true, Statement II is false.
4) Statement I is false, Statement II is true.

LEVEL-3

SINGLE CORRECT CHOICE TYPE:

13. The equation to the locus of a point P for which the distance from P to (0,5) is double the distance from P to y - axis is

1) $3x^2 + y^2 + 10y - 25 = 0$ 2) $3x^2 - y^2 + 10y + 25 = 0$

3) $3x^2 - y^2 + 10y - 25 = 0$ 4) $3x^2 + y^2 - 10y + 25 = 0$

14. The locus of P for which the distance from P to origin is double the distance from P to (1,2) is

1) $3x^2+3y^2-8x-16y+20=0$ 2) $3x^2+3y^2-8x+16y+20=0$

3) $3x^2-3y^2-8x-16y+20=0$ 4) $3x^2-3y^2-8x+16y+20=0$

MATRIX MATCH TYPE:

15.

COLUMN - I	COLUMN - II
a) The locus of the point $(at^2,2at)$ is	p) $xy = c^2$.
b) The locus of the point$(ct, c/t)$ is	q) $y^2+4x=4$
c) The locus of the point $(\cos^2 t, 2\sin t)$ is	r) $x^2+y^2=2$
d) The locus of the point $(\cos t+\sin t,\cos t-\sin t)$ is	s) $y^2= 4ax$

LEVEL-4

SINGLE CORRECT CHOICE TYPE:

16. If the distances from P to the points (5,-4), (7,6) are in the ratio 2 : 3, then the locus of P is

1) $5x^2+5y^2-12x-86y+17=0$ 2) $5x^2+5y^2-34x+120y+29=0$

3) $5x^2+5y^2-5x+y+14=0$ 4) $5x^2+5y^2+78x-40y+125=0$

17. A(2,3), B(2,-3) are two points. The equation to the locus of P such that PA +PB = 8 is

1) $16x^2+7y^2-64x-48=0$ 2) $16x^2+7y^2-64x+48=0$

3) $16x^2-7y^2+64x-48=0$ 4) $16x^2-7y^2+64x+48=0$

MATRIX MATCH TYPE:

18.

COLUMN - I	COLUMN - II
a) The locus of the point $\left(a\cos^4\theta, b\sin^4\theta\right)$ is	**p)** $\left(x^2y\right)^{2/3}+\left(xy^2\right)^{2/3}=1$
b) The locus of the point$\left(\tan\theta+\sin\theta,\tan\theta-\sin\theta\right)$ is	**q)** $x^2/a^2-y^2/b^2=1$
c) The locus of the point $(a\sec\theta, b\tan\theta)$ is	**r)** $\left(x^2-y^2\right)=16xy$
c) The colus of the point $(\operatorname{cosec}\theta-\sin\theta, \sec\theta-\cos\theta)$ is	**s)** $\sqrt{x/a}+\sqrt{y/b}=1$

LEVEL-5

SINGLE CORRECT CHOICE TYPE:

19. A(a,0), B(-a,0) are two points. The equation to the locus of P such that PA +PB = c is

1) $4\left(c^2-4a^2\right)x^2+4c^2y^2=c^2\left(c^2-4a^2\right)$ 2) $4\left(c^2+4a^2\right)x^2-4c^2y^2=c^2\left(c^2+4a^2\right)$

3) $2\left(c^2+2a^2\right)x^2+2c^2y^2=c^2\left(c^2+4a^2\right)$ 4) $2\left(c^2-4a^2\right)x^2-4c^2y^2=c^2\left(c^2-4a^2\right)$

20. A(ae,0), B(-ae,0) are two points. The equation to the locus of P such that PA - PB = 2a is

1) $\frac{x^2}{a^2}+\frac{y^2}{a^2\left(1-e^2\right)}=1$ 2) $\frac{x^2}{a^2}-\frac{y^2}{a^2\left(1-e^2\right)}=1$

3) $\frac{x^2}{a^2}+\frac{y^2}{a^2\left(1+e^2\right)}=1$ 4) $\frac{x^2}{a^2}-\frac{y^2}{a^2\left(1+e^2\right)}=1$

21. A(a.0) , B(-a,0) are two points . If a point P moves such that $\angle PAB-\angle PBA=\pi/2$ the locus of P is

1) $x^2+y^2=a^2$ 2) $x^2-y^2-a^2=0$ 3) $x^2-2xy-y^2=a^2$ 4) $b\left(x-p\right)=a\left(y-q\right)$

22. A(a,0), B(-a,0) are two points. If a point P moves such that $\angle PAB-\angle PBA=2\alpha$ then the locus of P is .

1) $x^2+2xy\,tan\,2\alpha-y^2=a^2$ 2) $x^2-2xy\,cot\,\alpha+y^2=a^2$

3) $x^2-2xy\,tan^2\,2\alpha+y^2=a^2$ 4) $x^2+2xy\,cot\,2\alpha-y^2=a^2$